

LEO POUILLY

See my projects in action ↓

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Engineering student passionate about robotics and mechanics in general. Actively developing my skills thanks to several projects that are closely related to the real world. My ultimate goal remains to create robots able to not only help humans in their professional life but also enhance their quality of life, fostering interaction between engineering, business, and global communities : a vision I am determined to pursue come rain or shine.

EDUCATION :

Cumulative GPA: 3.95/4.00

Arts et Métiers Institute of Technology (ENSAM)

Sept 2024 – Present

Combined BS & MS in Industrial & Mechanical Engineering

3.86/4.00 GPA

- **Top 10** French Graduate Engineering School
- **Relevant Coursework:** Mechanics (fluid, solids, continuum, finite element analysis), Materials Science, Thermodynamics, Mathematics (linear algebra, matrix, vector space), Computer Science, Industrial Simulation, Entrepreneurship, Product Design & Manufacturing, Industrial Organization

Lycée Hoche

Sept 2022 – Jul 2024

Elite undergraduate track in Advanced Mathematics & Physics ("Classe préparatoire aux grandes écoles")

4.00/4.00 GPA

- **Top 3** engineering preparatory school when I was admitted
- Intensive undergraduate program to prepare for highly competitive exams for admission to top Engineering Schools
- **Relevant Coursework:** Advanced mathematics (Algebra, Linear Algebra, analysis, topology) and Physics (mechanics, thermodynamics, electromagnetism)

Lycée Léonard de Vinci

Sept 2019 – Jul 2022

High School Diploma, Mathematics, Physics

4.00/4.00 GPA

- Graduated **top 5%** of class with **highest honors**

WORK AND LEADERSHIP EXPERIENCE

R&D Technical Contributor (Volunteer) - Carbon Risk Intelligence

Aug 2025 – Present

Helping my brother in his sustainable development startup by applying materials science skills to improve environmental datasets and observing startup operations

- Currently helping in classifying and interpreting data for CarbonCar, a SaaS that estimates the environmental footprint of automotive incidents
- Working with an **actuary** to understand risk modeling approaches used in the platform
- Observing how **due diligence** and **term sheet preparation** shape early client and investor discussions

Formula Student Arts & Métiers - FSAM Lille

Sept 2024 – Present

Objective: Work on **design** and optimization of FSAM Lille Formula Student **race car** preparing for European competitions

Team of 20

- **Corporate Relations Manager** : Lead **sponsorship strategy**, negotiate contracts, and manage partner relations securing over **€15,000** in funding
- **Powertrain Lead: Supervise engine design** and performance optimization. Coordinated a multidisciplinary subgroup to deliver **4 major powertrain components** from machining to integration, contributing to the car's core propulsion system

Business Manager - Student led business consultancy AMJE Lille

Jun 2025 – Present

Objective: Develop leadership, project management and customer relations skills while supervising engineering consultancy projects

- **Prospected 100+ potential corporate clients**, building a pipeline valued at **€20,000+ in prospective projects** for the association
- **Manage client relationships** and project timelines to ensure successful delivery of engineering solutions

Robotics in R&D intern - Les Companions - Industrial Startup automating construction tasks Jun 2025 – Aug 2025

Debugged, refined, and validated an untested prototype, enabling operation and integrating it into a robotic cell for industrial automation

- Tested and **optimized a robotic plastering end-effector**, redesigning CAD for durability and technical specifications
- Established measurement protocols to define optimal operational ranges for the system
- Integrated the improved prototype into a **robotic cell (UR5 robot)** to achieve full functional operation

Private Tutor — Mathematics, Physics

- Tutoring **Middle/High school students** in mathematics and physics, **adapting teaching** to individual needs
- Enhanced students' analytical skills and fostered measurable academic progress through personalized guidance

Les Entrep' Flandres — Top 5 Finalist

Oct. 2024 – Apr. 2025

Entrepreneurship Program

Team of 3

- Top 5 finish among 10+ teams in **France's largest student entrepreneurship program**
- Developed atmospheric water generator concept; assessed market via **BMC, PESTEL, SWOT and a Business Plan**
- Pitched business plan to judges, sharpening strategic thinking and communication skills

SELECTED RESEARCH & DEVELOPMENT PROJECTS

Hugging Face — LeRobot Project

Sept. 2025 – Present

Objective: Develop practical skills in robotics, AI control, and hardware/software integration

Individual Project

- Assembling and calibrating the **SO-101 robotic arm** to ensure precise electromechanical integration
- Leveraging the **LeRobot framework** to train manipulation policies via **Imitation Learning** and **Reinforcement Learning**
- Iteratively validating **control algorithms**, sensor feedback, and **motion planning** for **AI-driven autonomy**

ENEDIS Data Challenge — one-semester Academic Project

Sept. 2025 – Feb. 2026

Organized by *École Normale Supérieure (ENS) Ulm & Collège de France*

Team of 6

- Processed **200,000+ half-hourly load curves** from Linky smart meters to address real-world transmission gaps
- Built a benchmark comparing **interpolation methods** vs **ML models** (e.g., XGBoost/KNN) to impute missing values
- Optimized reconstruction accuracy (**RMSE**) on sparse time-series to enhance smart grid reliability

AI academic project

Sept 2024 – Jan 2025

Strengthened skills in machine learning, AI implementation, and problem-solving while mastering **Python, Keras, and**

Team of 2

Google Colab

- Trained **CNNs** (Convolutional Neural Networks) for supervised image classification on MNIST datasets
- Performed **feature extraction** at dense layers and applied **kNN** to evaluate predictive performance
- Applied **Transfer Learning** from MNIST to Fashion MNIST to improve training efficiency and accuracy

Research Project on Sports Betting Arbitrage (Source Code)

Nov 2022 – Jun 2024

Developed probabilistic and quantitative models to assess arbitrage opportunities in sports betting

Team of 2

- Modeled sports betting arbitrage using **sub-martingales** to identify systematic market patterns
- Performed **web scraping** to collect and structure odds data from **1,100+ matches across 300+ players**
- Analyzed **bookmaker parameters** and their statistical influence on odds fluctuations
- Evaluated arbitrage opportunities through **quantitative modeling** and statistical analysis

Personal Engineering Project - Custom Motorcycle (full rebuild & optimization)

Jun 2020 – Jun 2022

Applied mechanical, engine tuning, and integration skills to design a high-performance motorcycle

Individual Project

- Upgraded engine displacement with a **70cc Bidalot RF-WR** kit, replacing all internal components
- Increased power output **5× (from 4 CV to 20 CV)** by fine-tuning ignition and intake timing
- Installed and adapted **inverted R16V forks** and blue anodized supermoto wheels for improved handling

TECHNICAL & LANGUAGE SKILLS

Programming: Python (Selenium & BeautifulSoup, Qt, PyTorch, Scikit-learn, Pandas), C++, SQL, HTML/CSS, JavaScript

Software: 3DX, Catia, Abaqus, Onshape, Star CCM+(CFD), Matlab, Microsoft Office

Other: Data Analysis, Machine Learning, AI, LLM, Statistical modeling

French: Native

English: 93/120 TOEFL iBT

Spanish: Intermediate (B1)

German: Intermediate (B1)

Manual skills: 3D printing, CNC machining, Welding, Foundry

HOBBIES AND INTERESTS

Sports: Judo : Black belt at 15, multiple national training camps, Weight Training (2 times a week), Motocross(Leisure)

Judo Awards: Elected "Judoka of the Year" four times at Marck Judo, achieved multiple medals in **international judo tournaments** (minime category), **5th & 9th** place at the French **National Championship** (Minimes & Cadets)